Top 10 GO Biological Process Gene Sets Ranked by Potency of Perturbation (Sorted by BMD 25% Relative Deviation at the 5th Percentile)

DTTID: 108-020-004-000-5
Study Number:108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Heart

Date: 09 Jun 2025
Time: 09:35:54 AM

Female

Category Name	Input Genes/Platform Genes in Gene Set	% Gene Set Coverage	BMD_{rd25} at the 5th Percentile (ppm)	$BMDL_{rd25}$ at 5th Percentile - $BMDU_{rd25}$ at 5th Percentile (ppm)	Genes with Changed Direction Up	Genes with Changed Direction Down
GO:0044770 cell cycle phase transition	5/58	8.6	141.4	55.9 - 292.9	0	5
GO:0044772 mitotic cell cycle phase transition	5/52	9.6	141.4	55.9 - 292.9	0	5
GO:0051301 cell division	6/84	7.1	181.5	65.8 - 517.4	0	6
GO:0007049 cell cycle	4/45	8.9	192.5	68.7 - 382.0	0	4
GO:0000226 microtubule cytoskeleton organization	4/80	5.0	197.5	80.5 - 710.7	0	4
GO:0007059 chromosome segregation	4/41	9.8	202.9	85.1 - 551.4	0	4
GO:1903047 mitotic cell cycle process	11/149	7.4	242.4	96.3 - 435.9	0	11
GO:0022402 cell cycle process	12/194	6.2	282.0	108.6 - 1328.0	0	12
GO:0019730 antimicrobial humoral response	5/43	11.6	333.8	156.8 - 476.7	2	3
GO:0006959 humoral immune response	5/80	6.3	347.4	174.4 - 695.6	2	3

rd25 = 25% relative deviation

Active Genes:
GO:0044770: Usp22;Cks2;Ccnb2;Ccnb2-ps2;Ccna2
GO:0044772: Usp22;Cks2;Ccnb2;Ccnb2-ps2;Ccna2
GO:0051301: Nuf2;Cdca5;Kif18b;Ccnb2;Ccnb2-ps2;Ccna2
GO:0007049: Mki67;Cdca5;Kif18b;Cenpf
GO:0000226: Prc1;Nuf2;Kif18b;Ccnb2
GO:0007059: Smc2;Nuf2;Kif18b;Cenpf
GO:1903047: Usp22;Smc2;Prc1;Nuf2;Cdca5;Kif18b;Cks2;Ckap2;Ccnb2;Ccnb2-ps2;Ccna2
GO:0022402: Usp22;Smc2;Prc1;Nuf2;Cdca5;Kif18b;Cks2;Ckap2;Cenpf;Ccnb2;Ccnb2-ps2;Ccna2
GO:0019730: Loc102549061;Hist1h2bq;H2bc8;Ccl7;Ccl2
GO:0006959: Loc102549061;Hist1h2bq;H2bc8;Ccl7;Ccl2

Annotations
Official active gene symbols from the Rat Genome Database are shown in the "Active Genes" section above. Definitions of Gene Ontology terms were provided by the Gene Ontology Resource (http://geneontology.org/) and are included below.
GO process description version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

GO Biological Process Definitions:
GO:0044770 cell cycle phase transition: The cell cycle process by which a cell commits to entering the next cell cycle phase.
GO:0044772 mitotic cell cycle phase transition: The cell cycle process by which a cell commits to entering the next mitotic cell cycle phase.
GO:0051301 cell division: The process resulting in division and partitioning of components of a cell to form more cells; may or may not be accompanied by the physical separation of a cell into distinct, individually membrane-bounded daughter cells.
GO:0007049 cell cycle: The progression of biochemical and morphological phases and events that occur in a cell during successive cell replication or nuclear replication events. Canonically, the cell cycle comprises the replication and segregation of genetic material followed by the division of the cell, but in endocycles or syncytial cells nuclear replication or nuclear division may not be followed by cell division.
GO:0000226 microtubule cytoskeleton organization: A process that is carried out at the cellular level which results in the assembly, arrangement of constituent parts, or disassembly of cytoskeletal structures comprising microtubules and their associated proteins.

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GO:0007059 chromosome segregation: The process in which genetic material, in the form of chromosomes, is organized into specific structures and then physically separated and apportioned to two or more sets. In eukaryotes, chromosome segregation begins with the condensation of chromosomes, includes chromosome separation, and ends when chromosomes have completed movement to the spindle poles.
GO:1903047 mitotic cell cycle process: A process that is part of the mitotic cell cycle.
GO:0022402 cell cycle process: The cellular process that ensures successive accurate and complete genome replication and chromosome segregation.
GO:0019730 antimicrobial humoral response: An immune response against microbes mediated through a body fluid. Examples of this process are seen in the antimicrobial humoral response of Drosophila melanogaster and Mus musculus.
GO:0006959 humoral immune response: An immune response mediated through a body fluid.

** END OF REPORT **